7.1 Vectors in 2-Space

Introduction In science, mathematics, and engineering, we distinguish two important quantities: **scalars** and **vectors**. A scalar is simply a real number or a quantity that has *magnitude*. For example, length, temperature, and blood pressure are represented by numbers such as 80 m, 20°C, and the systolic/diastolic ratio 120/80. A vector, on the other hand, is usually described as a quantity that has both *magnitude* and *direction*.

Geometric Vectors Geometrically, a vector can be represented by a directed line segment—that is, by an arrow—and is denoted either by a boldface symbol or by a symbol with an arrow over it; for example, $\mathbf{v}$, $\vec{u}$, or $\overrightarrow{AB}$. Examples of vector quantities shown in FIGURE 7.1.1 are weight $\mathbf{w}$, velocity $\mathbf{v}$, and the retarding force of friction $\mathbf{F}_f$.

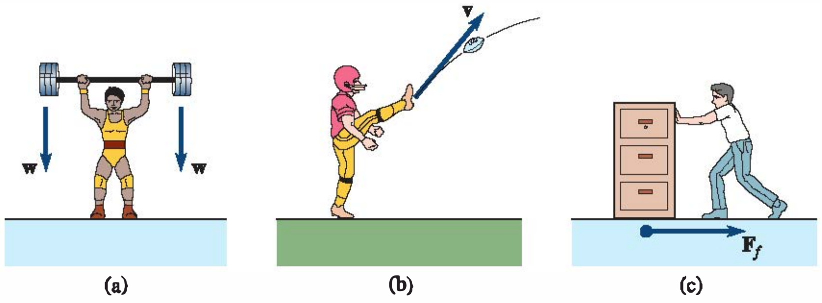


FIGURE 7.1.1 Examples of vector quantities

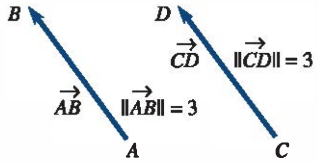


FIGURE 7.1.2 Vectors are equal

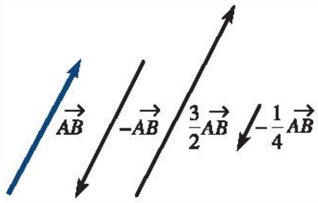


FIGURE 7.1.3 Parallel vectors

Notation and Terminology A vector whose initial point (or end) is A and whose terminal point (or tip) is B is written $\overrightarrow{AB}$. The magnitude of a vector is written $\|\overrightarrow{AB}\|$. Two vectors that have the same magnitude and same direction are said to be **equal**. Thus, in FIGURE 7.1.2, we have $\overrightarrow{AB} = \overrightarrow{CD}$. Vectors are said to be **free**, which means that a vector can be moved from one position to another provided its magnitude and direction are not changed. The **negative** of a vector $\overrightarrow{AB}$, written $-\overrightarrow{AB}$, is a vector that has the same magnitude as $\overrightarrow{AB}$ but is opposite in direction. If $k \neq 0$ is a scalar, the **scalar multiple** of a vector, $k\overrightarrow{AB}$, is a vector that is $|k|$ times as long as $\overrightarrow{AB}$. If $k > 0$, then $k\overrightarrow{AB}$ has the same direction as the vector $\overrightarrow{AB}$; if $k < 0$, then $k\overrightarrow{AB}$ has the direction opposite that of $\overrightarrow{AB}$. When $k = 0$, we say $0\overrightarrow{AB} = \mathbf{0}$ is the **zero vector**. Two vectors are **parallel** if and only if they are nonzero scalar multiples of each other. See FIGURE 7.1.3.

The question of what is the direction of $\mathbf{0}$ is usually answered by saying that the zero vector can be assigned *any* direction. More to the point, $\mathbf{0}$ is needed in order to have a vector algebra.

Addition and Subtraction Two vectors can be considered as having a common initial point, such as A in FIGURE 7.1.4(a). Thus, if nonparallel vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are the sides of a parallelogram as in Figure 7.1.4(b), we say the vector that is the main diagonal, or $\overrightarrow{AD}$, is the **sum** of $\overrightarrow{AB}$ and $\overrightarrow{AC}$. We write

$$\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{AC}.$$

The **difference** of two vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is defined by

$$\overrightarrow{AB} - \overrightarrow{AC} = \overrightarrow{AB} + (-\overrightarrow{AC}).$$

As seen in FIGURE 7.1.5(a), the difference $\overrightarrow{AB} - \overrightarrow{AC}$ can be interpreted as the main diagonal of the parallelogram with sides $\overrightarrow{AB}$ and $-\overrightarrow{AC}$. However, as shown in Figure 7.1.5(b), we can also interpret the same vector difference as the third side of a triangle with sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$. In this second interpretation, observe that the vector difference $\overrightarrow{CB} = \overrightarrow{AB} - \overrightarrow{AC}$ points toward the terminal point of the vector *from* which we are subtracting the second vector. If $\overrightarrow{AB} = \overrightarrow{AC}$, then

$$\overrightarrow{AB} - \overrightarrow{AC} = \mathbf{0}.$$

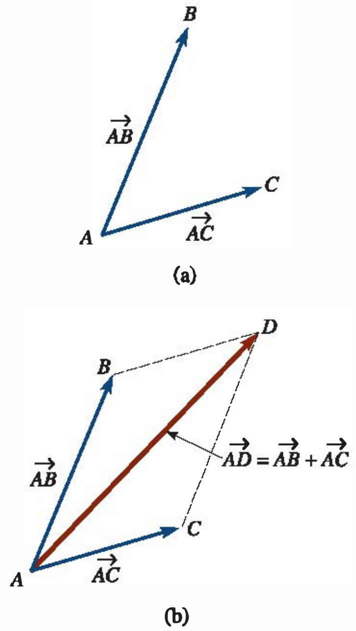
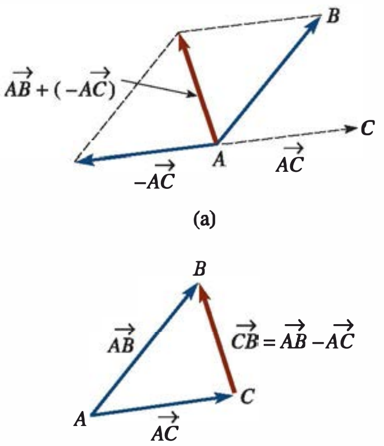


FIGURE 7.1.4 Vector $\overrightarrow{AD}$ is the sum of $\overrightarrow{AB}$ and $\overrightarrow{AC}$



(a)

(b)

FIGURE 7.1.5 Vector $\overrightarrow{CB}$ is the difference of $\overrightarrow{AB}$ and $\overrightarrow{AC}$

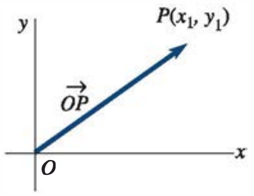
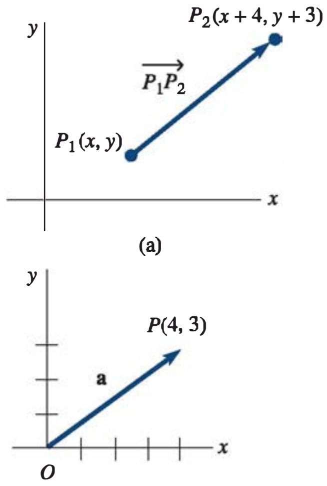


FIGURE 7.1.6 Position vector



(a)

(b)

FIGURE 7.1.7 Graphs of vectors in Example 1

Vectors in a Coordinate Plane To describe a vector analytically, let us suppose for the remainder of this section that the vectors we are considering lie in a two-dimensional coordinate plane or **2-space**. We shall denote the set of all vectors in the plane by R^2. The vector shown in **FIGURE 7.1.6**, with initial point the origin O and terminal point $P(x_1, y_1)$, is called the **position vector** of the point P and is written

$$\overrightarrow{OP} = \langle x_1, y_1 \rangle.$$

Components In general, a vector **a** in R^2 is any ordered pair of real numbers,

$$\mathbf{a} = \langle a_1, a_2 \rangle.$$

The numbers a_1 and a_2 are said to be the **components** of the vector **a**.

As we shall see in the first example, the vector **a** is not necessarily a position vector.

EXAMPLE 1 Position Vector

The displacement from the initial point $P_1(x, y)$ to the terminal point $P_2(x + 4, y + 3)$ in **FIGURE 7.1.7(a)** is 4 units to the right and 3 units up. As seen in Figure 7.1.7(b), the position vector of $\mathbf{a} = \langle 4, 3 \rangle$ emanating from the origin is equivalent to the displacement vector $\overrightarrow{P_1P_2}$ from $P_1(x, y)$ to $P_2(x + 4, y + 3)$. ≡

Addition and subtraction of vectors, multiplication of vectors by scalars, and so on, are defined in terms of their components.

Definition 7.1.1 Addition, Scalar Multiplication, Equality

Let $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$ be vectors in R^2.

(*i*) Addition: $\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2 \rangle$ (1)
(*ii*) Scalar multiplication: $k\mathbf{a} = \langle ka_1, ka_2 \rangle$ (2)
(*iii*) Equality: $\mathbf{a} = \mathbf{b}$ if and only if $a_1 = b_1, a_2 = b_2$ (3)

Subtraction Using (2), we define the **negative** of a vector **b** by

$$-\mathbf{b} = (-1)\mathbf{b} = \langle -b_1, -b_2 \rangle.$$

We can then define the **subtraction**, or the difference, of two vectors as

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}) = \langle a_1 - b_1, a_2 - b_2 \rangle. \qquad (4)$$

In **FIGURE 7.1.8(a)**, we have illustrated the sum of two vectors $\overrightarrow{OP_1}$ and $\overrightarrow{OP_2}$. In Figure 7.1.8(b), the vector $\overrightarrow{P_1P_2}$, with initial point P_1 and terminal point P_2, is the difference of position vectors

$$\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = \langle x_2 - x_1, y_2 - y_1 \rangle.$$

As shown in Figure 7.1.8(b), the vector $\overrightarrow{P_1P_2}$ can be drawn either starting from the terminal point of $\overrightarrow{OP_1}$ and ending at the terminal point of $\overrightarrow{OP_2}$, or as the position vector $\overrightarrow{OP}$ whose terminal point has coordinates $(x_2 - x_1, y_2 - y_1)$. Remember, $\overrightarrow{OP}$ and $\overrightarrow{P_1P_2}$ are considered equal, since they have the same magnitude and direction.

EXAMPLE 2 Addition and Subtraction of Two Vectors

If $\mathbf{a} = \langle 1, 4 \rangle$ and $\mathbf{b} = \langle -6, 3 \rangle$, find $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{b}$, and $2\mathbf{a} + 3\mathbf{b}$.

SOLUTION We use (1), (2), and (4):

$$\begin{aligned} \mathbf{a} + \mathbf{b} &= \langle 1 + (-6), 4 + 3 \rangle = \langle -5, 7 \rangle \\ \mathbf{a} - \mathbf{b} &= \langle 1 - (-6), 4 - 3 \rangle = \langle 7, 1 \rangle \\ 2\mathbf{a} + 3\mathbf{b} &= \langle 2, 8 \rangle + \langle -18, 9 \rangle = \langle -16, 17 \rangle. \end{aligned}$$ ≡

Properties

The component definition of a vector can be used to verify each of the following **properties** of vectors in R^2:

Theorem 7.1.1 Properties of Vectors

(i)	$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$	← commutative law
(ii)	$\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$	← associative law
(iii)	$\mathbf{a} + \mathbf{0} = \mathbf{a}$	← additive identity
(iv)	$\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$	← additive inverse
(v)	$k(\mathbf{a} + \mathbf{b}) = k\mathbf{a} + k\mathbf{b}$, k a scalar	
(vi)	$(k_1 + k_2)\mathbf{a} = k_1\mathbf{a} + k_2\mathbf{a}$, k_1 and k_2 scalars	
(vii)	$k_1(k_2\mathbf{a}) = (k_1k_2)\mathbf{a}$, k_1 and k_2 scalars	
(viii)	$1\mathbf{a} = \mathbf{a}$	
(ix)	$0\mathbf{a} = \mathbf{0}$	← zero vector

The **zero vector 0** in properties *(iii)*, *(iv)*, and *(ix)* is defined as

$$\mathbf{0} = \langle 0, 0\rangle.$$

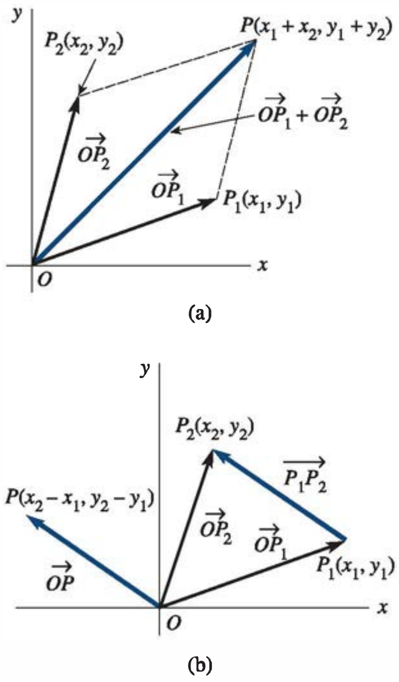


FIGURE 7.1.8 In (b), $\overrightarrow{OP}$ and $\overrightarrow{P_1P_2}$ are the same vector

Magnitude

The **magnitude**, **length**, or **norm** of a vector $\mathbf{a}$ is denoted by $\|\mathbf{a}\|$. Motivated by the Pythagorean theorem and **FIGURE 7.1.9**, we define the magnitude of a vector

$$\mathbf{a} = \langle a_1, a_2\rangle \quad \text{to be} \quad \|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2}.$$

Clearly, $\|\mathbf{a}\| \geq 0$ for any vector $\mathbf{a}$, and $\|\mathbf{a}\| = 0$ if and only if $\mathbf{a} = \mathbf{0}$. For example, if $\mathbf{a} = \langle 6, -2\rangle$, then $\|\mathbf{a}\| = \sqrt{6^2 + (-2)^2} = \sqrt{40} = 2\sqrt{10}$.

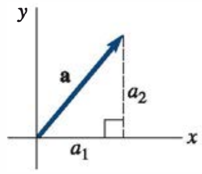


FIGURE 7.1.9 A right triangle

Unit Vectors

A vector that has magnitude 1 is called a **unit vector**. We can obtain a unit vector $\mathbf{u}$ in the same direction as a nonzero vector $\mathbf{a}$ by multiplying $\mathbf{a}$ by the positive scalar $k = 1/\|\mathbf{a}\|$ (reciprocal of its magnitude). In this case we say that $\mathbf{u} = (1/\|\mathbf{a}\|)\mathbf{a}$ is the **normalization** of the vector $\mathbf{a}$. The normalization of the vector $\mathbf{a}$ is a unit vector because

$$\|\mathbf{u}\| = \left\| \frac{1}{\|\mathbf{a}\|}\mathbf{a} \right\| = \frac{1}{\|\mathbf{a}\|}\|\mathbf{a}\| = 1.$$

Note: It is often convenient to write the scalar multiple $\mathbf{u} = (1/\|\mathbf{a}\|)\mathbf{a}$ as

$$\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|}.$$

EXAMPLE 3 Unit Vectors

Given $\mathbf{a} = \langle 2, -1\rangle$, form a unit vector in the same direction as $\mathbf{a}$. In the opposite direction of $\mathbf{a}$.

SOLUTION The magnitude of the vector $\mathbf{a}$ is $\|\mathbf{a}\| = \sqrt{2^2 + (-1)^2} = \sqrt{5}$. Thus a unit vector in the same direction as $\mathbf{a}$ is the scalar multiple

$$\mathbf{u} = \frac{1}{\sqrt{5}}\mathbf{a} = \frac{1}{\sqrt{5}}\langle 2, -1\rangle = \left\langle \frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}} \right\rangle.$$

A unit vector in the opposite direction of $\mathbf{a}$ is the negative of $\mathbf{u}$:

$$-\mathbf{u} = \left\langle -\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right\rangle.$$

≡

If $\mathbf{a}$ and $\mathbf{b}$ are vectors and c_1 and c_2 are scalars, then the expression $c_1\mathbf{a} + c_2\mathbf{b}$ is called a **linear combination** of $\mathbf{a}$ and $\mathbf{b}$. As we see next, any vector in R^2 can be written as a linear combination of two special vectors.

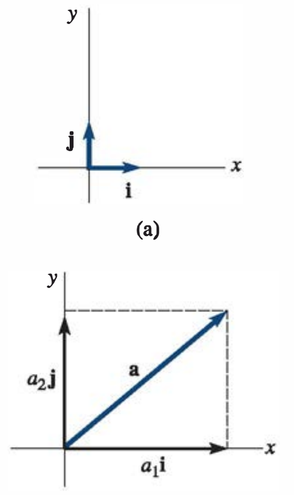


FIGURE 7.1.10 i and j form a basis for R^2

The i, j Vectors

In view of (1) and (2), any vector $\mathbf{a} = \langle a_1, a_2 \rangle$ can be written as a sum:

$$\langle a_1, a_2 \rangle = \langle a_1, 0 \rangle + \langle 0, a_2 \rangle = a_1\langle 1, 0 \rangle + a_2\langle 0, 1 \rangle. \tag{5}$$

The unit vectors $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$ are usually given the special symbols **i** and **j**. See **FIGURE 7.1.10(a)**. Thus, if

$$\mathbf{i} = \langle 1, 0 \rangle \quad \text{and} \quad \mathbf{j} = \langle 0, 1 \rangle,$$

then (5) becomes

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}. \tag{6}$$

The unit vectors **i** and **j** are referred to as the **standard basis** for the system of two-dimensional vectors, since any vector **a** can be written uniquely as a linear combination of **i** and **j**. If $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$ is a position vector, then Figure 7.1.10(b) shows that **a** is the sum of the vectors $a_1\mathbf{i}$ and $a_2\mathbf{j}$, which have the origin as a common initial point and which lie on the x- and y-axes, respectively. The scalar a_1 is called the **horizontal component** of **a**, and the scalar a_2 is called the **vertical component** of **a**.

EXAMPLE 4 Vector Operations Using i and j

(a) $\langle 4, 7 \rangle = 4\mathbf{i} + 7\mathbf{j}$
(b) $(2\mathbf{i} - 5\mathbf{j}) + (8\mathbf{i} + 13\mathbf{j}) = 10\mathbf{i} + 8\mathbf{j}$
(c) $\|\mathbf{i} + \mathbf{j}\| = \sqrt{2}$
(d) $10(3\mathbf{i} - \mathbf{j}) = 30\mathbf{i} - 10\mathbf{j}$
(e) $\mathbf{a} = 6\mathbf{i} + 4\mathbf{j}$ and $\mathbf{b} = 9\mathbf{i} + 6\mathbf{j}$ are parallel, since **b** is a scalar multiple of **a**. We see that $\mathbf{b} = \frac{3}{2}\mathbf{a}$. ≡

EXAMPLE 5 Graphs of Vector Sum/Vector Difference

Let $\mathbf{a} = 4\mathbf{i} + 2\mathbf{j}$ and $\mathbf{b} = -2\mathbf{i} + 5\mathbf{j}$. Graph $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$.

SOLUTION The graphs of $\mathbf{a} + \mathbf{b} = 2\mathbf{i} + 7\mathbf{j}$ and $\mathbf{a} - \mathbf{b} = 6\mathbf{i} - 3\mathbf{j}$ are given in **FIGURE 7.1.11(a)** and 7.1.11(b), respectively.

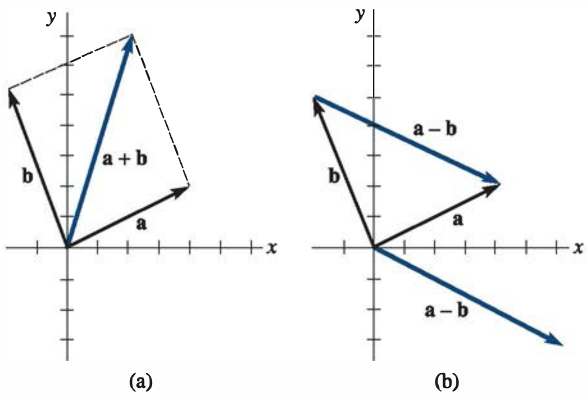


FIGURE 7.1.11 Graphs of vectors in Example 5

≡

7.1 Exercises

Answers to selected odd-numbered problems begin on page ANS-14.

In Problems 1–8, find (a) $3\mathbf{a}$, (b) $\mathbf{a} + \mathbf{b}$, (c) $\mathbf{a} - \mathbf{b}$, (d) $\|\mathbf{a} + \mathbf{b}\|$, and (e) $\|\mathbf{a} - \mathbf{b}\|$.

1. $\mathbf{a} = 2\mathbf{i} + 4\mathbf{j}, \ \mathbf{b} = -\mathbf{i} + 4\mathbf{j}$
2. $\mathbf{a} = \langle 1, 1 \rangle, \ \mathbf{b} = \langle 2, 3 \rangle$
3. $\mathbf{a} = \langle 4, 0 \rangle, \ \mathbf{b} = \langle 0, -5 \rangle$
4. $\mathbf{a} = \frac{1}{6}\mathbf{i} - \frac{1}{6}\mathbf{j}, \ \mathbf{b} = \frac{1}{2}\mathbf{i} + \frac{5}{6}\mathbf{j}$
5. $\mathbf{a} = -3\mathbf{i} + 2\mathbf{j}, \ \mathbf{b} = 7\mathbf{j}$
6. $\mathbf{a} = \langle 1, 3 \rangle, \ \mathbf{b} = -5\mathbf{a}$